Fair Reconstruction Error for SPDMD with and without Mean Subtraction

Internal note — Project 1

1 SPDMD Reconstruction in Reduced Space

The snapshot matrix after preprocessing (optional mean subtraction and/or linear detrending) is

$$\boldsymbol{D} = \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_{m-1} \end{bmatrix} \in \mathbb{R}^{n\times(m-1)}, \tag{1}$$

where n is the spatial dimension and m the total number of snapshots.

The economy SVD of $\boldsymbol{D}$ (truncated to rank r) gives

$$\boldsymbol{D} \approx \boldsymbol{U}_r\, \boldsymbol{\Sigma}_r\, \boldsymbol{V}_r^{\top}, \tag{2}$$

where $\boldsymbol{U}_r \in \mathbb{R}^{n\times r}$, $\boldsymbol{\Sigma}_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r) \in \mathbb{R}^{r\times r}$, $\boldsymbol{V}_r \in \mathbb{R}^{(m-1)\times r}$. In the code, `SVD_S` stores $\boldsymbol{\Sigma}_r$ and `SVD_VT` stores $\boldsymbol{V}_r^{\top}$.

1.1 Reduced data matrix $\boldsymbol{G}$

The data projected into the r-dimensional reduced space is

$$\boldsymbol{G} = \boldsymbol{\Sigma}_r\, \boldsymbol{V}_r^{\top} \;\in\; \mathbb{R}^{r\times(m-1)}. \tag{3}$$

This is computed in `dmd.f95` as

```
G = matmul(transpose(SVD_S), transpose(SVD_VT))
```

Note that $\boldsymbol{G} = \boldsymbol{U}_r^{\top}\boldsymbol{D}$, i.e. the projection of $\boldsymbol{D}$ onto its left singular vectors.

1.2 DMD eigenvalues and the Vandermonde matrix

After computing the reduced dynamics matrix $\tilde{\boldsymbol{A}} = \boldsymbol{U}_r^{\top}\boldsymbol{D}''\boldsymbol{V}_r\boldsymbol{\Sigma}_r^{-1}$ and solving its eigenvalue problem $\tilde{\boldsymbol{A}}\boldsymbol{W} = \boldsymbol{W}\boldsymbol{\Lambda}$, the DMD eigenvalues $\mu_1, \ldots, \mu_r$ define the Vandermonde matrix

$$\boldsymbol{C} \;\in\; \mathbb{C}^{r\times(m-1)}, \qquad C_{j,k} = \mu_j^{k-1}, \quad j = 1, \ldots, r, \;\; k = 1, \ldots, m-1. \tag{4}$$

1.3 SPDMD sparse amplitudes

The sparsity-promoting optimisation (Jovanović et al. 2014) solves, for each penalty γ,

$$\min_{\boldsymbol{b}} \; \underbrace{\|\boldsymbol{G} - \boldsymbol{W}\;\mathrm{diag}(\boldsymbol{b})\,\boldsymbol{C}\|_{\mathrm{F}}^2}_{J_1} + \gamma\;\|\boldsymbol{b}\|_1\,, \tag{5}$$

where $\boldsymbol{W} = \boldsymbol{V}_r$ (the DMD eigenvectors in the reduced basis, stored as `Vr`) and $\boldsymbol{b} \in \mathbb{C}^r$ are the mode amplitudes. The matrices

$$\boldsymbol{P} = \big(\boldsymbol{W}^H\boldsymbol{W}\big) \circ \overline{(\boldsymbol{C}\,\boldsymbol{C}^H)} \;\in\; \mathbb{C}^{r\times r}, \tag{6}$$

$$\boldsymbol{q} = \mathrm{diag}\big(\boldsymbol{W}^H\,\boldsymbol{G}\,\boldsymbol{C}^H\big) \;\in\; \mathbb{C}^r, \tag{7}$$

(where $\circ$ denotes Hadamard product) reduce the least-squares term to

$$J_1 = \boldsymbol{b}^H \boldsymbol{P}\, \boldsymbol{b} - 2\,\mathrm{Re}\big(\boldsymbol{q}^H \boldsymbol{b}\big) + \|\boldsymbol{G}\|_{\mathrm{F}}^2 \,. \tag{8}$$

The ADMM solver (`amp.f95`, module `Sparity.f95`) produces sparse amplitudes $\boldsymbol{b}_\gamma$ for each γ value.

2 Reconstruction Error

Given sparse amplitudes $\boldsymbol{b}_\gamma$ with `nzval` nonzero entries, the reduced-space reconstruction is

$$\hat{\boldsymbol{G}}_\gamma = \mathrm{Re}(\boldsymbol{W}\ \mathrm{diag}(\boldsymbol{b}_\gamma)\, \boldsymbol{C}) \ \in\ \mathbb{R}^{r\times(m-1)}. \tag{9}$$

This is computed in `reconstruction_opt_error.f95` as:

```
B(j,j) = B_p(j, r4)       ! diagonal matrix of sparse amplitudes
xdmd = real(matmul(matmul(Vr, B), C))
```

The reconstruction error is

$$\boxed{\epsilon(\gamma) = \frac{\left\|\boldsymbol{G} - \hat{\boldsymbol{G}}_\gamma\right\|_{\mathrm{F}}^2}{\|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2} \times 100\%.} \tag{10}$$

The denominator $\|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2 = \sum_{i=1}^{r} \sigma_i^2$ equals the total energy captured by the rank-r truncation. In the code:

```
F_norm_orgi = sum(SVD_S**2)        ! = ||Sigma_r||_F^2
F_norm_XDMD = sum((G_matrix - xdmd)**2)  ! = ||G - G_hat||_F^2
error = (F_norm_XDMD / F_norm_orgi) * 100
```

3 The Denominator Difference: `mean0` vs. `mean1`

The crucial observation is that **the SVD is performed on different data** depending on whether the temporal mean is subtracted:

- `mean0` (no mean subtraction): $\boldsymbol{D}^{(0)} = \boldsymbol{X}$ (raw snapshots $\to$ SVD gives $\sigma_i^{(0)}$).
- `mean1` (mean subtracted): $\boldsymbol{D}^{(1)} = \boldsymbol{X} - \bar{\boldsymbol{x}}\,\boldsymbol{1}^\top$ (fluctuations $\to$ SVD gives $\sigma_i^{(1)}$).

Since $\boldsymbol{D}^{(0)}$ contains the mean field, its singular values are larger:

$$\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2 \ >\ \left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2 . \tag{11}$$

The respective reconstruction errors are

$$\epsilon^{(0)} = \frac{\left\|\boldsymbol{G}^{(0)} - \hat{\boldsymbol{G}}^{(0)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2} \times 100\%, \tag{12}$$

$$\epsilon^{(1)} = \frac{\left\|\boldsymbol{G}^{(1)} - \hat{\boldsymbol{G}}^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2} \times 100\%. \tag{13}$$

These use **different denominators**, so comparing $\epsilon^{(0)}$ to $\epsilon^{(1)}$ directly is not meaningful.

4 Fair Error Metric

To compare reconstruction quality on the same footing, we normalise $\epsilon^{(1)}$ so that both errors use the *same denominator* — the total energy including the mean:

$$\alpha \;=\; \frac{\left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2} \;=\; \frac{\sum_{i=1}^{r}\left(\sigma_i^{(1)}\right)^2}{\sum_{i=1}^{r}\left(\sigma_i^{(0)}\right)^2}. \tag{14}$$

This is the **fluctuation-to-total energy ratio**. Then the fair reconstruction error for `mean1` is

$$\boxed{\epsilon_{\text{fair}}^{(1)} \;=\; \alpha \cdot \epsilon^{(1)} \;=\; \frac{\left\|\boldsymbol{G}^{(1)} - \hat{\boldsymbol{G}}^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2} \times 100\%.} \tag{15}$$

For `mean0`, the fair error equals the original error:

$$\epsilon_{\text{fair}}^{(0)} \;=\; \epsilon^{(0)}. \tag{16}$$

4.1 Derivation

The scaling is simply a change of denominator:

$$\begin{aligned}
\epsilon_{\text{fair}}^{(1)} &= \alpha \cdot \epsilon^{(1)} \\
&= \frac{\left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2} \cdot \frac{\left\|\boldsymbol{G}^{(1)} - \hat{\boldsymbol{G}}^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2} \times 100\% \\
&= \frac{\left\|\boldsymbol{G}^{(1)} - \hat{\boldsymbol{G}}^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2} \times 100\%.
\end{aligned} \tag{17}$$

The $\left\|\boldsymbol{\Sigma}_r^{(1)}\right\|_{\mathrm{F}}^2$ cancels, leaving the reconstruction residual of the fluctuation field normalised by the total energy. Both $\epsilon_{\text{fair}}^{(0)}$ and $\epsilon_{\text{fair}}^{(1)}$ are now fractions of the same quantity.

4.2 Exactness of α: no truncation error

A potential concern is whether α is affected by SVD truncation. In general, if the SVD were truncated to some rank $r < m-1$, the sums $\sum_{i=1}^{r}\sigma_i^2$ would discard the tail of the spectrum and α would only be an approximation.

In our implementation, however, the SVD rank equals $m-1$ in all cases (verified across all 39 HDF5 result files). Since $\boldsymbol{D} \in \mathbb{R}^{n\times(m-1)}$ with $n \gg m-1$, the matrix has at most $m-1$ nonzero singular values, and the economy SVD with $r = m-1$ captures **all of them**. Therefore

$$\|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2 = \sum_{i=1}^{m-1}\sigma_i^2 = \|\boldsymbol{D}\|_{\mathrm{F}}^2\,, \tag{18}$$

and the energy ratio is *exact*:

$$\alpha = \frac{\left\|\boldsymbol{D}^{(1)}\right\|_{\mathrm{F}}^2}{\left\|\boldsymbol{D}^{(0)}\right\|_{\mathrm{F}}^2} = \frac{\text{total fluctuation energy}}{\text{total energy (incl. mean)}}. \tag{19}$$

No singular values are discarded, so α involves no truncation approximation whatsoever.

Note that the sparsity parameter γ only affects the *numerator* of the error, i.e. how many DMD modes are used to approximate $\boldsymbol{G}$. The *denominator* $\|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2$ always uses all $m-1$ singular values regardless of how many modes are retained. Alpha simply re-scales this denominator to be the same for both `mean0` and `mean1`.

4.3 Physical interpretation of α

- $\alpha \approx 1$: The mean carries little energy; the data is fluctuation-dominated. Fair and raw errors are similar. *Example:* Eulerian gas fraction, where $\alpha \approx 0.82$–0.92.
- $\alpha \ll 1$: The mean dominates the total energy. The raw `mean1` error $\epsilon^{(1)}$ is inflated (normalised by small fluctuation energy), making `mean0` *appear* much better. *Example:* Moving-Eulerian gas fraction, where $\alpha \approx 0.03$–0.31; Lagrangian gas fraction $\alpha \approx 0.03$–0.08.

5 Why the Naïve Comparison is Misleading

Consider Lagrangian gas fraction, Case C_1 (Re20 Bo20):

Variant	ϵ (raw)	α	ϵ_{fair}
`mean0`	0.04%	0.026	0.04%
`mean1`	4.33%	0.026	0.11%

Naïvely, `mean0` appears $\sim 100\times$ better (0.04% vs. 4.33%). After fair correction, `mean0` is only $\sim 3\times$ better (0.04% vs. 0.11%) — and this residual difference is on fluctuation energy that is only 2.6% of the total, i.e. both variants reconstruct $> 99.9\%$ of the total field energy.

The naïve comparison is misleading because the `mean0` denominator $\left\|\boldsymbol{\Sigma}_r^{(0)}\right\|_{\mathrm{F}}^2$ includes the mean energy (which is trivially reproduced), inflating the denominator and shrinking the percentage.

6 Summary

1. The standard SPDMD error metric $\epsilon = \left\|\boldsymbol{G} - \hat{\boldsymbol{G}}\right\|_{\mathrm{F}}^2 / \|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2 \times 100\%$ uses $\boldsymbol{\Sigma}_r$ from whichever data was decomposed.
2. When comparing `mean0` and `mean1`, the denominators differ: `mean0` includes mean energy, `mean1` does not.
3. The fair error $\epsilon_{\mathrm{fair}}^{(1)} = \alpha\,\epsilon^{(1)}$ normalises both by the same total energy.
4. After fair correction, `mean0` and `mean1` produce **comparable** reconstruction quality for non-Eulerian frames. For Eulerian frames, `mean1` is genuinely much better.
5. Mean subtraction is numerically **essential** for SPDMD in non-Eulerian frames (prevents Vandermonde overflow and ADMM failure) and costs nothing in reconstruction accuracy.

7 Manuscript–Code Consistency Check

This section verifies that the theoretical formulation in the manuscript (Section 2, "Data-driven modal decomposition methods") is consistent with the Fortran implementation, and provides a complete variable mapping.

7.1 Variable mapping: manuscript $\longleftrightarrow$ code

Manuscript	Code variable	File	Description
$\mathcal{D}'$	`u_stacked_full(:,1:m-1)`	`dmd.f95`	First $m-1$ snapshots cessing (mean/trend s
$\mathcal{D}''$	Built from snapshots $2..m$	`dmd.f95`	Shifted snapshot matr
U_r	`SVD_U` $(n \times r)$	`dmd.f95`	Left singular vectors
Σ_r	`SVD_S` $(r \times r)$	`dmd.f95`	Diagonal matrix of sin
W_r (manuscript)	`SVD_VT` $(m-1 \times r)$	`dmd.f95`	Right singular vectors **Note:** code stores V_r, the name
$\tilde{A} = U_r^T \mathcal{D}'' W_r \Sigma_r^{-1}$	`A` $\rightarrow$ `A_comp` $(r \times r)$	`dmd.f95`	Reduced dynamics ma
γ_j (manuscript Eq. 8)	`Mu2`, `Mu`	`dmd.f95`	DMD eigenvalues μ_j
$\tilde{v}_j$	`Vr` $(r \times r)$ from `zgeev`	`dmd.f95`	Eigenvectors of $\tilde{A}$
$\varphi_j = U_r \tilde{v}_j$	`Phi` $(n_c \times r)$	`recon_opt_memory.f95`	Full-space DMD mod reconstruction)
V_γ (Vandermonde)	`Vand`, `C` $(r \times m-1)$	`dmd.f95`, `recon_opt_error.f95`	$C_{j,k} = \mu_j^{k-1}$
$B_b = \mathrm{diag}(\boldsymbol{b})$	`B` $= \mathrm{diag}($`B_p(:,`γ`_idx))`	`recon_opt_error.f95`	Diagonal sparse ampli
—	`G` $= \Sigma_r V_r^T$ $(r \times m-1)$	`dmd.f95`	Reduced data mat manuscript)

7.2 Notation issue: eigenvalue symbol clash

The manuscript uses γ_j for the eigenvalues of $\tilde{A}$ (Eq. 8). However, in the SPDMD literature (Jovanović et al. 2014) and in the code, γ is the **sparsity penalty parameter**, while the eigenvalues are denoted μ_j. This creates a notational conflict: the SPDMD optimisation is

$$\min_{\boldsymbol{b}} \|\boldsymbol{G} - \boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{C}\|_{\mathrm{F}}^2 + \gamma\,\|\boldsymbol{b}\|_1\,,$$

where γ is the penalty weight and μ_j are the eigenvalues that build the Vandermonde matrix $\boldsymbol{C}$. **Recommendation:** Use μ_j for DMD eigenvalues throughout the manuscript to avoid ambiguity with the SPDMD penalty γ.

7.3 Derivation: from manuscript (full-space) to code (reduced-space)

The manuscript states the SPDMD optimisation problem as (paraphrased):

$$\min_{\boldsymbol{b}} \left\|\mathcal{D}' - \boldsymbol{\varphi}\, B_b\, V_\gamma\right\|_{\mathrm{F}}^2. \tag{20}$$

This is in **full space** $(n_s \times (m-1))$. The code instead works in the **reduced r-dimensional space**. We now show these are mathematically equivalent.

Step 1: Substitute $\boldsymbol{\varphi} = U_r \boldsymbol{W}$. From the manuscript (Eq. 9), the DMD spatial modes are $\varphi_j = U_r \tilde{v}_j$, so $\boldsymbol{\varphi} = U_r \boldsymbol{W}$, where $\boldsymbol{W} \in \mathbb{C}^{r\times r}$ holds the eigenvectors of $\tilde{A}$ as columns (code variable `Vr`). Thus

$$\mathcal{D}' - \boldsymbol{\varphi}\, B_b\, V_\gamma = \mathcal{D}' - U_r\, \boldsymbol{W}\, B_b\, \boldsymbol{C}. \tag{21}$$

Step 2: Express $\mathcal{D}'$ in the SVD basis. From the truncated SVD (Eq. 7): $\mathcal{D}' = U_r \boldsymbol{\Sigma}_r V_r^T + \boldsymbol{E}_{\text{trunc}}$, where $\boldsymbol{E}_{\text{trunc}}$ is the truncation residual. Since $r = m-1$ in our case, $\boldsymbol{E}_{\text{trunc}} = \boldsymbol{0}$ and we have exactly:

$$\mathcal{D}' = U_r \underbrace{\boldsymbol{\Sigma}_r V_r^T}_{=\boldsymbol{G}}. \tag{22}$$

Step 3: Use orthonormality of U_r. Since $U_r^T U_r = I_r$, the Frobenius norm is preserved under left multiplication by U_r:

$$\begin{aligned} \left\|\mathcal{D}' - U_r \boldsymbol{W} B_b \boldsymbol{C}\right\|_{\mathrm{F}}^2 &= \|U_r \boldsymbol{G} - U_r \boldsymbol{W} B_b \boldsymbol{C}\|_{\mathrm{F}}^2 \\ &= \|U_r\,(\boldsymbol{G} - \boldsymbol{W} B_b \boldsymbol{C})\|_{\mathrm{F}}^2 \\ &= \|\boldsymbol{G} - \boldsymbol{W}\, B_b\, \boldsymbol{C}\|_{\mathrm{F}}^2\,. \end{aligned} \tag{23}$$

The last step holds because for any matrix $\boldsymbol{X}$ and orthonormal U_r:

$$\|U_r \boldsymbol{X}\|_{\mathrm{F}}^2 = \operatorname{tr}\!\left(\boldsymbol{X}^T U_r^T U_r \boldsymbol{X}\right) = \operatorname{tr}\!\left(\boldsymbol{X}^T \boldsymbol{X}\right) = \|\boldsymbol{X}\|_{\mathrm{F}}^2\,.$$

Conclusion. The full-space problem (20) is **exactly equivalent** to the reduced-space problem solved by the code:

$$\boxed{\min_{\boldsymbol{b}}\ \|\boldsymbol{G} - \boldsymbol{W}\ \operatorname{diag}(\boldsymbol{b})\, \boldsymbol{C}\|_{\mathrm{F}}^2 \ =\ \min_{\boldsymbol{b}}\ \left\|\mathcal{D}' - \boldsymbol{\varphi}\ \operatorname{diag}(\boldsymbol{b})\, V_\gamma\right\|_{\mathrm{F}}^2.} \tag{24}$$

This equivalence holds exactly because $r = m-1$ (no SVD truncation). Even if $r < m-1$, the optimisation would still be equivalent (the truncation error $\|\boldsymbol{E}_{\text{trunc}}\|_{\mathrm{F}}^2$ is constant w.r.t. $\boldsymbol{b}$ and drops out of the arg min), though the error *values* would differ by that constant.

7.4 Reconstruction error: code formula

The reconstruction in reduced space (code `reconstruction_opt_error.f95`):

$$\hat{\boldsymbol{G}} = \operatorname{Re}(\boldsymbol{W}\ \operatorname{diag}(\boldsymbol{b}_\gamma)\, \boldsymbol{C})\,, \tag{25}$$

$$\epsilon = \frac{\left\|\boldsymbol{G} - \hat{\boldsymbol{G}}\right\|_{\mathrm{F}}^2}{\|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2} \times 100\%. \tag{26}$$

Using the equivalence (23), this is identical to the full-space relative error:

$$\epsilon = \frac{\left\|\mathcal{D}' - \hat{\mathcal{D}}'\right\|_{\mathrm{F}}^2}{\|\mathcal{D}'\|_{\mathrm{F}}^2} \times 100\%, \tag{27}$$

where $\hat{\mathcal{D}}' = U_r \hat{\boldsymbol{G}} = \boldsymbol{\varphi}\ \operatorname{diag}(\boldsymbol{b}_\gamma)\, V_\gamma$ and the denominator uses

$$\left\|\mathcal{D}'\right\|_{\mathrm{F}}^2 = \left\|U_r \boldsymbol{\Sigma}_r V_r^T\right\|_{\mathrm{F}}^2 = \|\boldsymbol{\Sigma}_r\|_{\mathrm{F}}^2 = \sum_{i=1}^{r} \sigma_i^2.$$

Code implementation (lines 288–298 of `reconstruction_opt_error.f95`):

```
xdmd = real(matmul(matmul(Vr, B), C))       ! G_hat = Re(W * diag(b) * C)

summ = 0.0_dp
do i = 1, r1
    do j = 1, m-1
        summ = summ + (G_matrix(i,j) - xdmd(i,j))**2
    end do
end do
F_norm_XDMD = summ                           ! ||G - G_hat||_F^2

error = (F_norm_XDMD / F_norm_orgi) * 100    ! F_norm_orgi = sum(SVD_S**2)
```

7.5 The SPDMD matrices $\boldsymbol{P}$ and $\boldsymbol{q}$: expanding the quadratic

The manuscript states that "using SVD properties, the definition of matrix φ, and some algebra, the result is then in a quadratic form that is solved using ADMM." Here we show this algebra explicitly, connecting to Jovanović et al. (2014).

Expanding the reduced-space objective:

$$\begin{aligned} J(\boldsymbol{b}) &= \|\boldsymbol{G} - \boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{C}\|_{\mathrm{F}}^2 \\ &= \mathrm{tr}\big[(\boldsymbol{G} - \boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{C})^H(\boldsymbol{G} - \boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{C})\big] \\ &= \|\boldsymbol{G}\|_{\mathrm{F}}^2 - 2\,\mathrm{Re}(\boldsymbol{q}^H\boldsymbol{b}) + \boldsymbol{b}^H\boldsymbol{P}\,\boldsymbol{b}, \end{aligned} \tag{28}$$

where

$$P_{ij} = (\boldsymbol{W}^H\boldsymbol{W})_{ij}\,\overline{(\boldsymbol{C}\boldsymbol{C}^H)_{ij}}, \tag{29}$$

$$q_i = \Big[\mathrm{diag}\Big(\overline{\boldsymbol{C}\,\boldsymbol{G}^T\,\boldsymbol{W}}\Big)\Big]_i. \tag{30}$$

These are the Hadamard (element-wise) products from Jovanović et al. (2014, Eq. 11–12). In `dmd.f95` (lines ∼1033–1044):

```
P = matmul(conjg(transpose(Vr)), Vr) &          ! W^H * W
    * conjg(matmul(Vand, conjg(transpose(Vand)))) ! .* conj(C * C^H)

G = matmul(transpose(SVD_S), transpose(SVD_VT))  ! Sigma_r * V_r^T

q_diag = conjg(matmul(matmul(Vand, transpose(G)), Vr))  ! conj(C * G^T * W)
do i = 1, rank
    q(i) = q_diag(i,i)                           ! take diagonal
end do
```

This matches Eqs. (29)–(30) exactly. The ADMM solver in `Sparity.f95` then minimises $J(\boldsymbol{b}) + \gamma\,\|\boldsymbol{b}\|_1$ over $\boldsymbol{b}$.

7.6 Verdict: manuscript vs. code

Item	Match?	Comment
SVD of $\mathcal{D}'$	✓	Manuscript Eq. 7 = code `dgesvd`.
$\tilde{A} = U_r^T \mathcal{D}'' W_r \Sigma_r^{-1}$	✓	Manuscript Eq. 8 = code `A = temp_two * Inv_Sigma`. SVD variable naming mismatch: manuscript calls right singular vectors W_r, code names them `SVD_VT` (actually stores V_r).
Eigenvalue problem	✓	Manuscript Eq. 8 = code `zgeev(A_comp,...)`.
Spatial modes $\varphi_j = U_r \tilde{v}_j$	✓	Manuscript Eq. 9 = code `Phi`.
SPDMD optimisation	✓	Manuscript (full-space) $\equiv$ code (reduced-space) via U_r orthonormality.
Reconstruction	✓	$\hat{\boldsymbol{G}} = \mathrm{Re}(\boldsymbol{W}\,\mathrm{diag}(\boldsymbol{b})\boldsymbol{C})$ matches code exactly.
Error denominator	✓	$\lVert \boldsymbol{\Sigma}_r \rVert_{\mathrm{F}}^2 = \lVert \mathcal{D}' \rVert_{\mathrm{F}}^2$ (exact since $r = m-1$).
Eigenvalue notation	×	Manuscript uses γ_j; code/SPDMD literature uses μ_j. Conflicts with SPDMD penalty γ.

Bottom line: The manuscript's theoretical formulation and the code implementation are **mathematically equivalent**. The manuscript describes the problem in full space (n_s-dimensional), while the code works in reduced space (r-dimensional) — a standard and exact simplification when U_r has orthonormal columns. The only issue is the notation γ_j for eigenvalues, which should be changed to μ_j to avoid confusion with the SPDMD sparsity parameter γ.